

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2– Concept Mapping (5 QUESTIONS)

Course Code:	ITA05	Course Title:	Computer Vision
CO Assessed:	CO1 – Imitation (BL3)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	30% of CIA	Total Marks:	50 (5 Questions × 10Marks)
CO1 Statement:	Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems.		
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Section / Batch:	Ai-MI-2024	Date:	16/07/26

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Code of Conduct

I, B. Rakesh (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: 

Instructions

- This test contains 5 questions, each carrying 10 marks. Total: 50 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Overview of Computer Vision, Features of Computer Vision, and Levels of Computer Vision	Computer Vision pipeline, features, and processing levels	1	10
Digital Image Fundamentals	Image representation, resolution, and image quality	2	10
Image Sensing and Acquisition, Sampling and Quantization	Digital image acquisition, sampling, and quantization process	3	10
Image Formation Models	Real-world scene to digital image formation	4	10
Applications of Image Processing and Computer Vision	Computer Vision capabilities and real-world applications	5	10
GRAND TOTAL:			50 Marks

Annexure A –Concept mapping

1. Construct a comprehensive and well-organized concept map illustrating the complete pipeline from real-world scene capture to digital image representation. Identify all major concepts, establish meaningful relationships among intermediate transformations and dependencies, integrate insights from at least five recent literature sources, and present the map with clear hierarchy, linking phrases, and logical organization.

2. Develop a hierarchically structured concept map illustrating the major stages of visual data handling in a Computer Vision system. Clearly show the relationships among acquisition, preprocessing, feature extraction, analysis, and interpretation stages, explain their contributions to meaningful outputs, incorporate findings from at least five literature sources, and ensure clarity and readability.

3. Create a detailed concept map explaining the relationships among image resolution, spatial resolution, intensity resolution, and image quality. Clearly illustrate cause-and-effect relationships, discuss their impact on image usability in different applications, integrate evidence from at least five research articles, and organize the concepts logically.

4. Design an analytically organized concept map illustrating the factors influencing image acquisition conditions, including illumination, sensor characteristics, motion, environment, and optics. Show how these factors affect downstream image processing and application performance, integrate relevant literature findings, and present the map in a visually coherent manner.

5. Construct an original concept map illustrating the relationship between Computer Vision system capabilities and the selection of appropriate vision-based applications. Identify major system capabilities, establish their influence on application effectiveness, integrate perspectives from multiple literature sources, and present the concepts using a clear hierarchical structure.

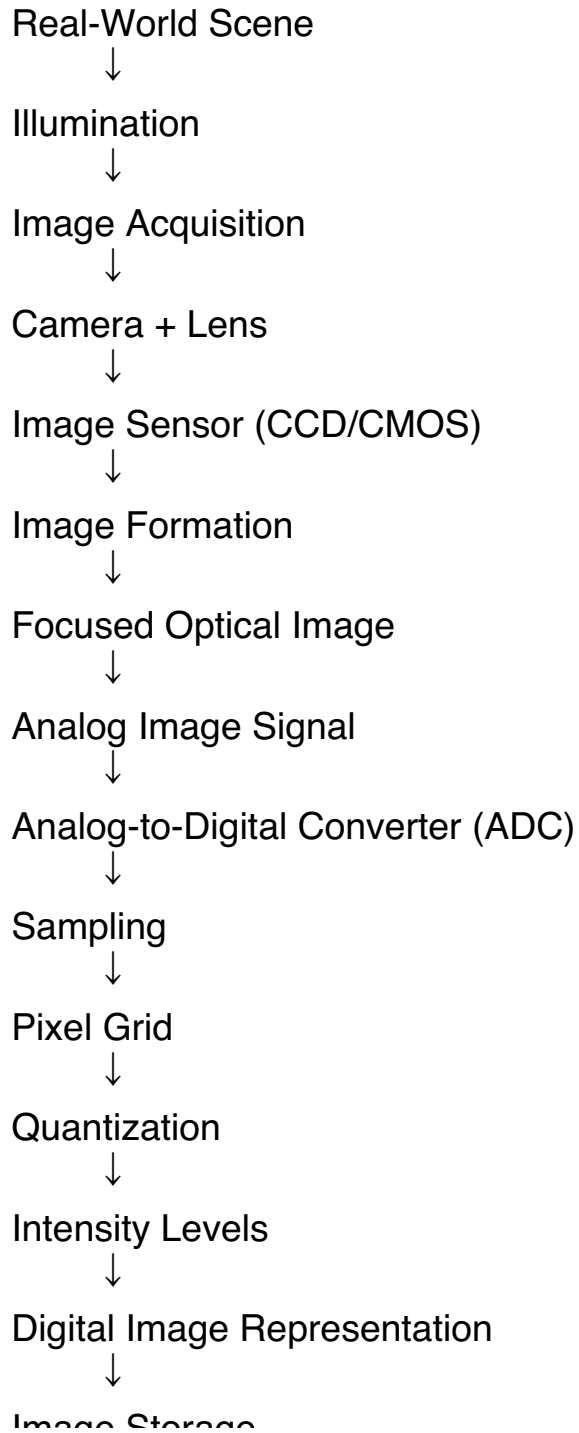
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Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks	
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts		
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established		
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure		
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence		
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand		
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q1. From Real-World Scene to Digital Image Representation



Literature keywords

Gonzalez & Woods

Richard Szeliski

OpenCV Documentation

Sonka et al.

Forsyth & Po

Q2. Major Stages of Visual Data Handling

Real-World Scene



Image Acquisition



Pre-processing



Noise Removal



Image Enhancement



Segmentation



Feature Extraction



Feature Selection



Object Detection



Classification



Recognition



Interpretation



Applications:

Face Recognition

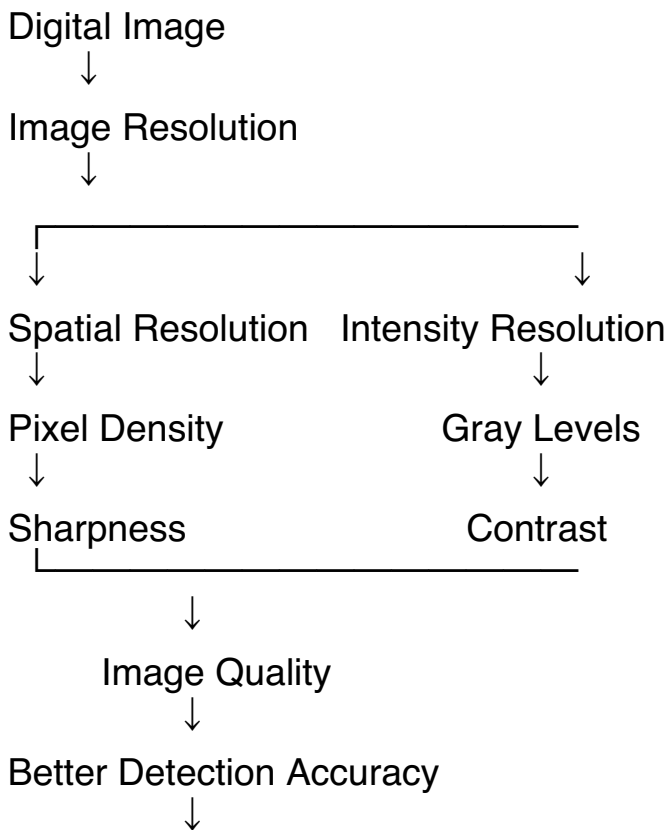
Medical Diagnosis

Traffic Monitoring

Agriculture

Robotics

Q3.Relationship Among Resolution and Image Quality



Examples:

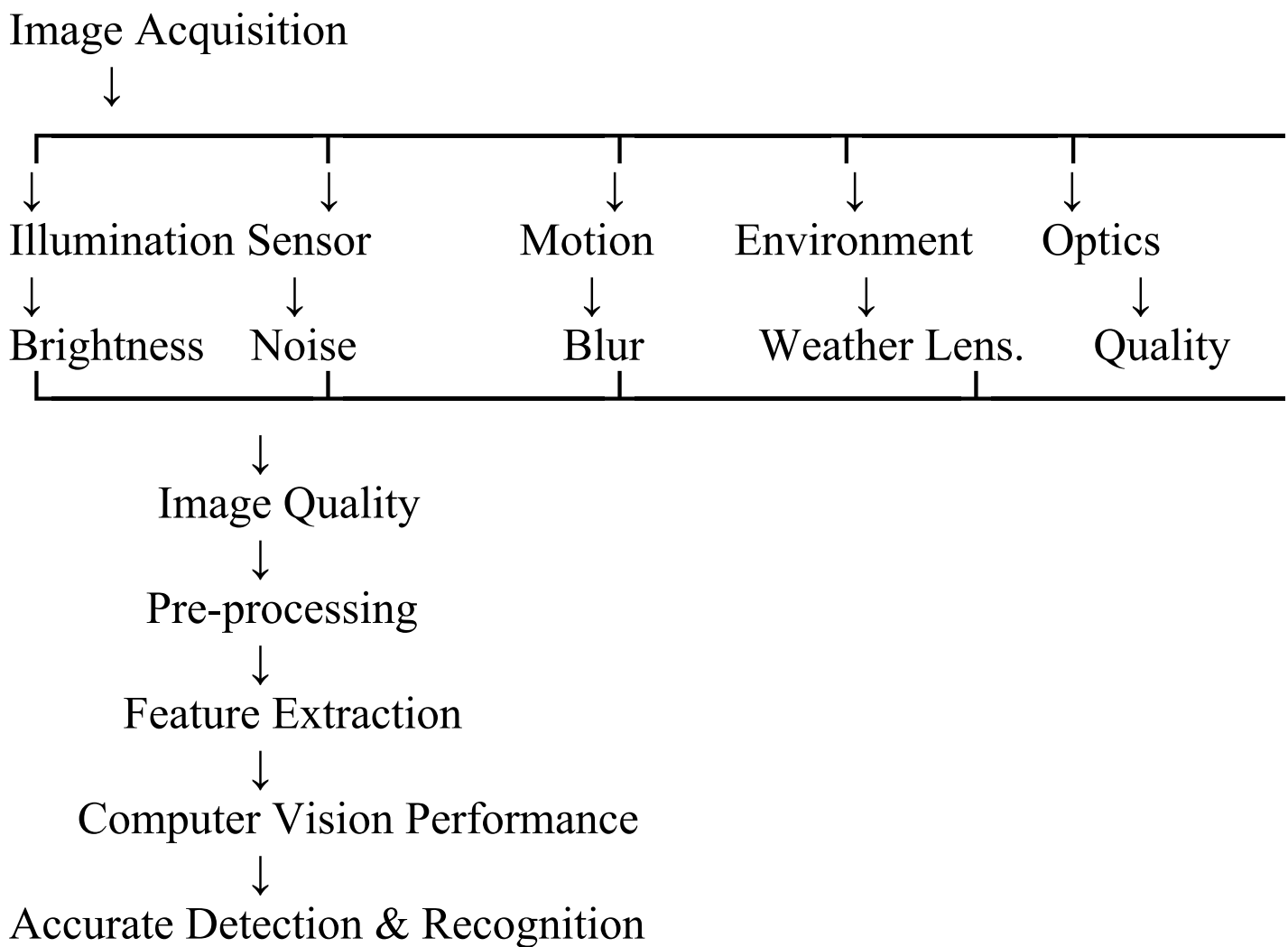
Medical Imaging

Satellite Images

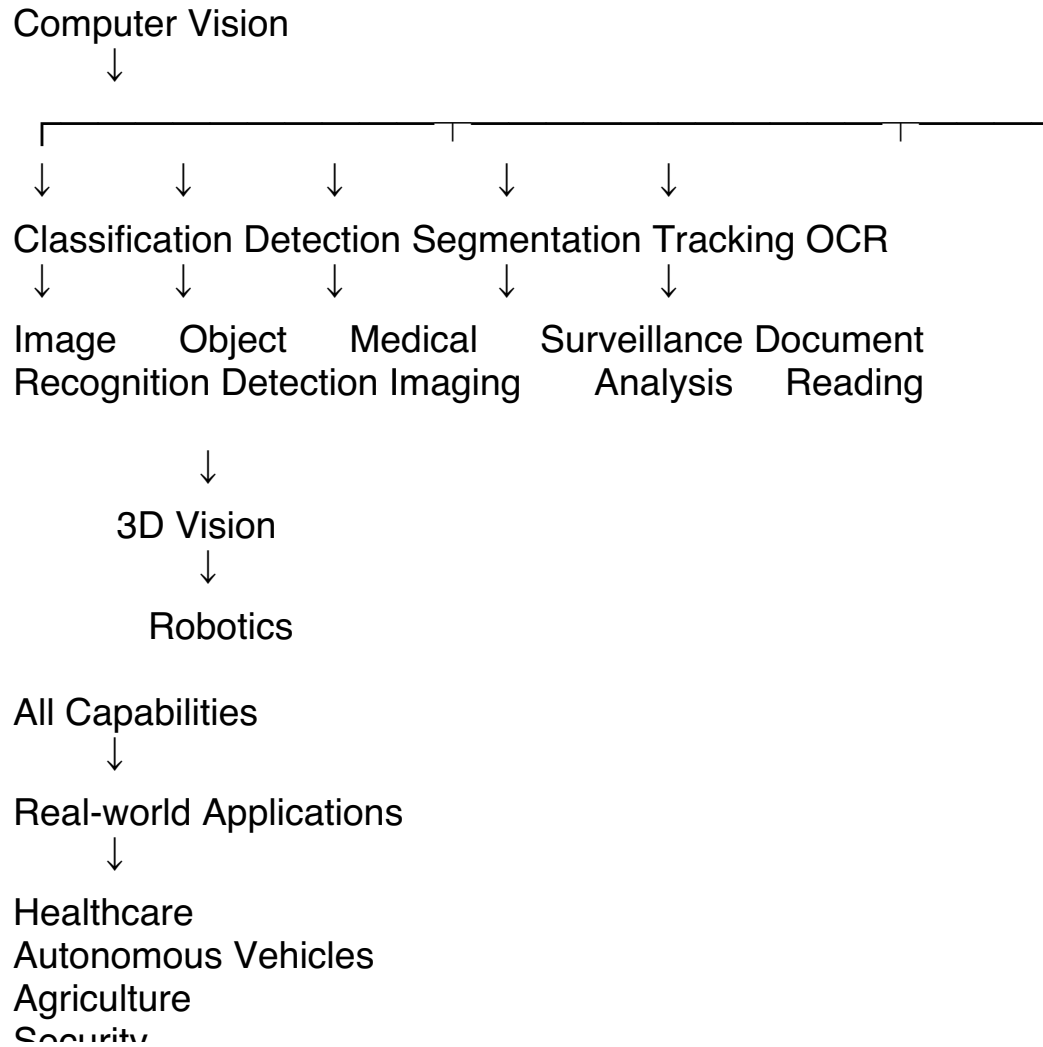
Face Recognition

Industrial Inspection

Q4. Factors Influencing Image Acquisition



Q5. Computer Vision Capabilities and Applications



Literature Sources

Gonzalez & Woods – Digital Image Processing

Richard Szeliski – Computer Vision: Algorithms and Applications

OpenCV Documentation

Sonka, Hlavac & Boyle – Image Processing, Analysis and Machine Vision

Forsyth & Ponce – Computer Vision: A Modern Approach